

SANA: LakeQA SUBSET20B results

Experiment records 2026-08-31-web-arm-subset20b and 2026-08-31-model-tiers-subset20b

1 Results

1.1 Retrieval source: open web versus the data lake

To test whether the retrieval axis reflects the data lake specifically or a more general property of the agent, we replaced lake search with open-web search. The arm uses the Parallel Search API and, under `--no-s3`, loses every lake-backed file tool: the agent keeps only `download`, which fetches an arbitrary `http(s)` URL into its sandbox, and `execute_code`. All arms hold the remaining axes at `search_results=naive`, `profile=standard` and `computation_tool=standard`, since the ideal computation tools resolve against authored data-lake records and would make a web run’s outcome independent of what it retrieved. Each arm was run three times.

The web arm reaches 38.3% against 56.7% for the strongest lake arm, a gap of 18.3pp that clears the ± 12 pp two-sigma replicate threshold measured across these arms.

The more informative quantity is where retrieval breaks down. D_{ret} is the fraction of gold datasets that appeared in search results; D_{acc} is the fraction the agent actually read. All three lake arms surface gold at a similar rate (Ideal 82.5%, Standard 84.1%, Naive 85.6%), so they are not separated by what search returns. They are separated by the drop between finding and reading (Ideal 3.4pp, Standard 16.2pp, Naive 15.7pp): oracle search wins because the agent acts on what it was shown, not because it was shown more.

For the web arm both quantities are undefined rather than zero: it touches no lake datasets by construction, so it can neither retrieve nor read a gold source.

Table 1: Retrieval-source comparison on LakeQA SUBSET20B (gpt-5-mini, 20 tasks per arm, three replicate rounds, semantic match). $\bar{x}$ is the mean over rounds and σ their standard deviation. *Blank* counts answers the judge classified as honest abstentions rather than wrong answers. *Cap* is the mean number of tasks per round reaching the 30-call tool budget. D_{ret} and D_{acc} are retrieval and access recall of gold datasets, undefined for the web arm.

Arm	R1	R2	R3	$\bar{x}$ (%)	σ	Blank	Cap	D_{ret} (%)	D_{acc} (%)
Ideal	50	55	65	56.7	7.6	2	6.0	82.5	79.1
Standard	50	50	55	51.7	2.9	0	10.7	84.1	67.9
Naive	45	65	60	56.7	10.4	0	9.7	85.6	69.9
Web	35	40	40	38.3	2.9	5	7.7	n/a	n/a

What the web arm actually measures. `download` accepts any URL, and the model routinely constructs one from pretrained knowledge rather than following a search result: live Socrata queries against `data.cityofnewyork.us`, `data.va.gov` export endpoints, and a timestamped `web.archive.org` URL. The arm therefore measures an agent with open web access and code execution, not web-search retrieval, and its data path may be stronger than the lake arms’ since it aggregates live authoritative sources rather than a fixed snapshot. That liveness also produces false negatives: two failures were arithmetically correct against current data but disagreed with gold answers derived from the snapshot (2,700,968 against a gold 2,705,988; 2923.59 against 2941).

1.2 Replicate variance on the ablation grid

We ran the leave-one-out ablation over three model tiers on SUBSET20B, a 20-task split of LakeQA stratified jointly on node count and historical solve rate. Every axis is held at the oracle and one axis is degraded per cell, giving seven cells per model. Each cell was run **three times** under identical configuration, so every number carries a standard deviation rather than resting on a single sample.

That replication is the central result. The mean per-cell standard deviation is 7.1 pp, so a two-sigma interval spans ± 14 pp and an effect must exceed roughly 14 pp to be distinguishable from run-to-run variance. Since one task is worth 5 pp at $n = 20$, that threshold is about 3 tasks.

Of the 18 degraded cells, only 5 clear that threshold: Search: BM25 at **gpt-5.4-nano** (-25.0 pp), Plan: Default at **gpt-5.2** (-20.0 pp), Search: BM25 at **gpt-5-mini** (-16.7 pp), Search: BM25 at **gpt-5.2** (-15.0 pp), Search: PNEUMA at **gpt-5-mini** (-15.0 pp). The search axis is the one effect that replicates across every tier; the remaining cells cannot be separated from noise at this sample size.

The reference cells themselves are **gpt-5.4-nano** 41.7% ($\sigma = 2.9$), **gpt-5-mini** 73.3% ($\sigma = 2.9$), **gpt-5.2** 71.7% ($\sigma = 10.4$). **gpt-5-mini** and **gpt-5.2** differ by 1.7 pp against a standard error of 6.2 pp on that difference, so they are not separable here. On this split the frontier model does not outperform the smaller one, and a single round would have suggested otherwise: round 1 alone put **gpt-5.2** 5 pp below **gpt-5-mini**.

Table 2: Leave-one-out ablation across model tiers on LakeQA SUBSET20B (20 tasks per cell, semantic match, three replicate rounds under identical configuration). $\bar{x}$ is the mean over rounds and σ their standard deviation; δ is the effect relative to that model’s reference cell. Effects exceeding the ± 14 pp two-sigma threshold are set in bold.

Model	Condition	R1	R2	R3	$\bar{x}$ (%)	σ	δ (pp)
gpt-5.4-nano	Reference (all ideal)	40	45	40	41.7	2.9	—
	Search: BM25	25	20	5	16.7	10.4	-25.0
	Search: PNEUMA	35	35	20	30.0	8.7	-11.7
	Search: Preloaded	50	45	35	43.3	7.6	$+1.7$
	Plan: No Plan	30	45	40	38.3	7.6	-3.3
	Plan: Default	40	40	15	31.7	14.4	-10.0
	Data An.: Standard	40	25	35	33.3	7.6	-8.3
gpt-5-mini	Reference (all ideal)	70	75	75	73.3	2.9	—
	Search: BM25	55	65	50	56.7	7.6	-16.7
	Search: PNEUMA	50	60	65	58.3	7.6	-15.0
	Search: Preloaded	85	75	75	78.3	5.8	$+5.0$
	Plan: No Plan	65	60	60	61.7	2.9	-11.7
	Plan: Default	65	60	65	63.3	2.9	-10.0
	Data An.: Standard	65	75	65	68.3	5.8	-5.0
gpt-5.2	Reference (all ideal)	60	80	75	71.7	10.4	—
	Search: BM25	60	55	55	56.7	2.9	-15.0
	Search: PNEUMA	65	65	50	60.0	8.7	-11.7
	Search: Preloaded	85	75	80	80.0	5.0	$+8.3$
	Plan: No Plan	75	65	60	66.7	7.6	-5.0
	Plan: Default	60	45	50	51.7	7.6	-20.0
	Data An.: Standard	75	55	55	61.7	11.5	-10.0

Table 3: Replicate noise floor, over every cell run three times. One task is worth 5 pp at $n = 20$, so the mean per-cell standard deviation is close to a single task and the two-sigma threshold is about 3.

Statistic over per-cell σ	Value (pp)
Mean	7.1
Median	7.6
Maximum	14.4
Two-sigma threshold	14.1
Cells	21

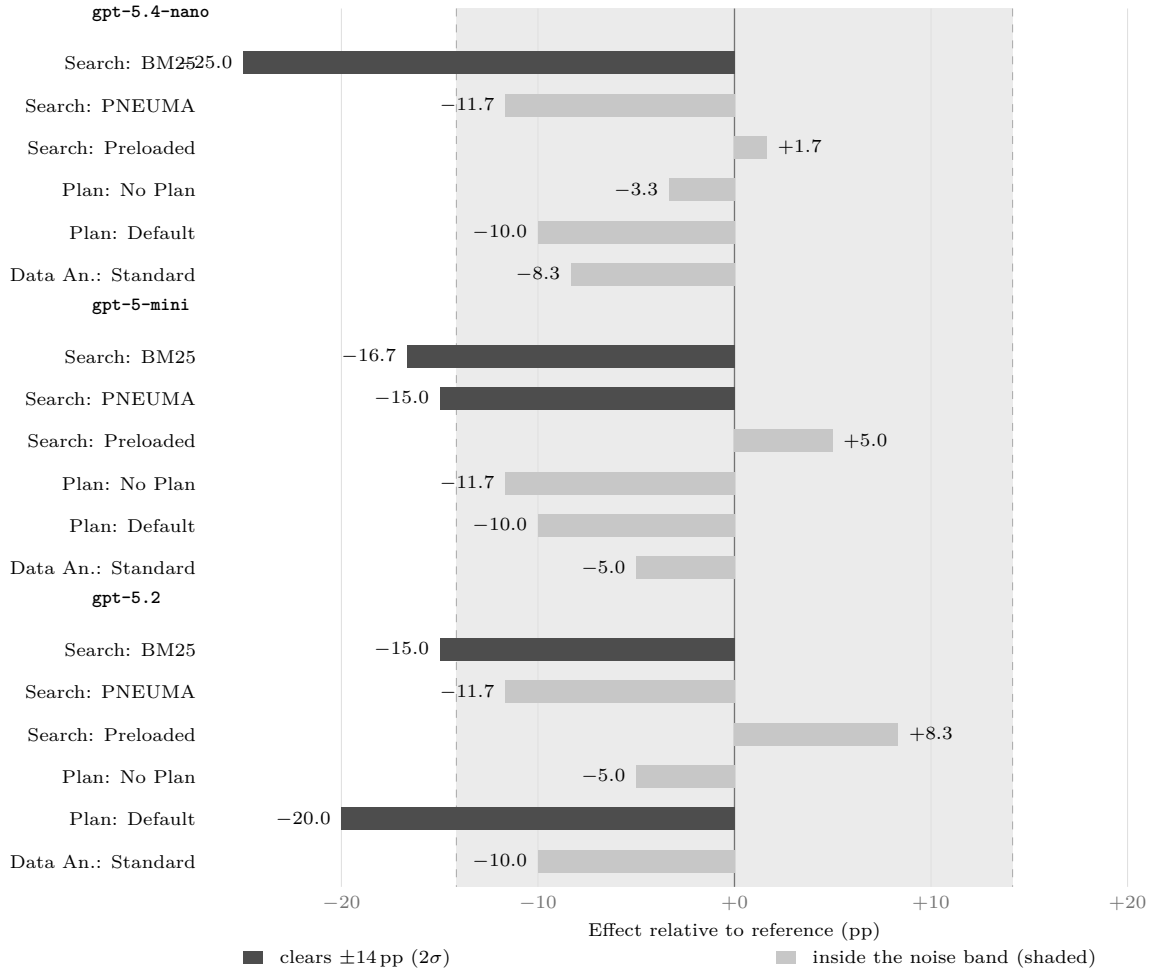


Figure 1: Ablation effects across three replicate rounds. Each bar is the change from that model’s all-ideal reference when one axis is degraded; the shaded column is the ± 14 pp two-sigma noise band. Bars ending inside it cannot be distinguished from run-to-run variance at $n = 20$, where one task is worth 5 pp.